Summary in Graph

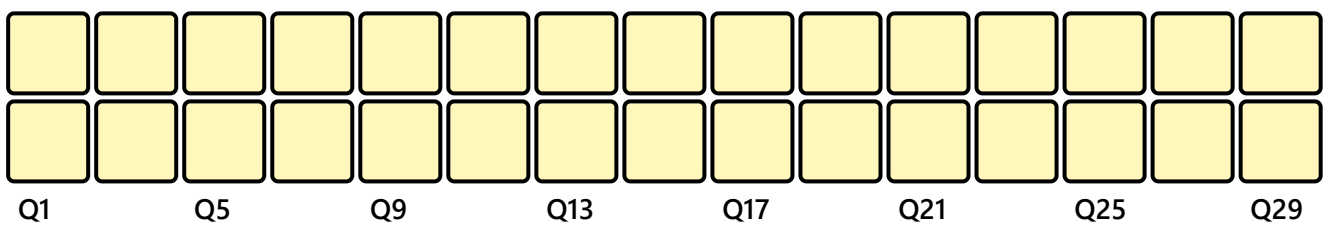
Exam Summary (GO Classes CS Test Series 2026 | Computer Networks | Subject Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (25 + 5)
Total Marks:	35 (25 + 10)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (28) out of 275 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

An organization is assigned the address block $14.24.74.0/24$. The first subnet is allocated 128 addresses (for 120 users) using a / 25 prefix. What is the first address of the next available subnet (e.g., for 60 hosts)?

A. 14.24.74.64
B. 14.24.74.126

C. 14.24.74.128
D. 14.24.74.127

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

From the address block $140.25.0.0, a/27$ subnet is created, If only 25 of these addresses are used or assigned, what percentage of the subnet's address space is not utilised?

A. 12.5%
B. 21.875%
C. 6.25%
D. 25%

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Assume that a $12-bit$ Hamming codeword consisting of $8-bit$ data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$, where the data bits and the check bits are given in the following tables:

d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1
1	1	0	x	0	1	0	1

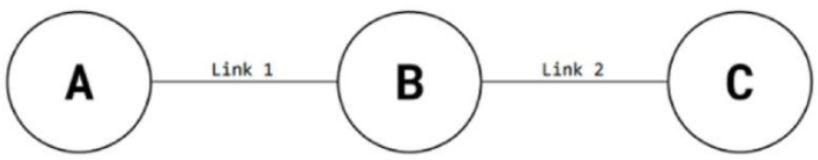
c_8	c_4	c_2	c_1
y	0	1	0

Assuming even parity, find the value of $xXNORy$_____.

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Consider the diagram below. Link 1 has length L_1 m (where m stands for meters) and allows packets to be propagated at speed S_1 m/sec, while Link 2 has length L_2 m but it only allows packets to be propagated at speed S_2 m/sec (because two links are made of different materials). Link 1 has transmission rate T_1 bits/sec and Link 2 has transmission rate T_2 bits /sec. At time 0 , Node A sends packet P_1 with D_1 Bytes and then it sends another packet



P_2 with D_2 Bytes immediately after it pushes all bits of P_1 onto Link 1 .
Find the variable relations that need to be satisfied in order to have no queuing delays.

A. $\frac{D_1}{T_2} \leq \frac{D_2}{T_1}$
B. $\frac{D_1}{T_2} \geq \frac{D_2}{T_1}$
C. $T_2 \geq T_1$

D. $D_2 \geq D_1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Consider the following three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links.
(S1.) The computational overhead in link state protocols is higher than in distance vector 2 protocols.
(S2.) A distance vector protocol (with split horizon) avoids persistent routing loops, but not a } link state protocol.
(S3.) After a topology change, a link state protocol will converge faster than a distance vector protocol.

Which one of the following is correct ?

A. S1
B. S2
C. S3
D. None of them are correct.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

$$\text{CWND} = \text{CWND bytes}$$
$$\text{MSS} = MSS \text{ bytes}$$

In CA phase, at each Acknowledgment, how many bytes CWND should increase (assuming we get Acknowledgment for complete segment and each seg is Acknowledged)

A. $\text{CWND}+ = \frac{\text{CWND}}{\text{MSS}}$
B. $\text{CWND}+ = \frac{\text{MSS}}{\text{CWND}}$
C. $\text{CWND}+ = \frac{\text{MSS}\times\text{MSS}}{CWND}$
D. $\text{CWND}+ = \frac{\text{CWND}}{\text{MSS}}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

A. HTTP GET request, DNS query, TCP SYN
B. DNS query, TCP SYN, HTTP GET request.
C. DNS query, HTTP GET request, TCP SYN
D. TCP SYN, DNS query, HTTP GET request.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

Assume that the bandwidth for a TCP connection is 1048560 bits $/\text{sec}$. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are

A. 63 milliseconds, 65535×2^{14}
B. 63 milliseconds, 65535×2^{16}
C. 500 milliseconds, 65535×2^{14}
D. 500 milliseconds, 65535×2^{16}

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

The persistent timer is used in TCP to
(I.) To detect crashes from the other end of the connection
(II.) To enable retransmission
(III.) To avoid deadlock condition
(IV.) To timeout FIN Wait1 condition

Which of the above statements are True?

A. (i) and (iii) only
B. (i) and (ii) only
C. (i), (ii), and (iii) only
D. (iii) only

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #10 Numerical Type Award: 1 Penalty: 0 Computer Networks

A computer on a 10 Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2 Mbps . It is initially filled to capacity with 16 Megabits . What is the maximum duration for which the computer can transmit at the full 10 Mbps (in seconds)

Your Answer: Correct Answer: 2 Not Attempted Time taken: 00min 00sec Discuss

Q #11 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

Given that the requested information is not available at local DNS server, a purely recursive DNS query from a requesting host would follow the path:

A. Root server, Local server, TLD server, Local server, Authoritative server.
B. Root server, Authoritative server, TLD server.
C. Root server, TLD server, Authoritative server.
D. Root server, Local server, Authoritative server, Local server, TLD server.

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Which of the following application layer protocol(s) is/are push protocol?
1 SMTP
2 IMAP
3 POP3
4 HTTP

A. 1 only
B. 2 and 3
C. 4 only
D. 3 only

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A company is assigned three IP address blocks: $149.16.23.0/22$, $149.16.24.0/22$, and $149.16.30.0/22$. It wants to divide these blocks into 12 equal-sized subnets for internal use. What subnet mask should be assigned to each of the 12 subnets?

A. $/22$
B. $/23$
C. $/24$
D. $/25$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Suppose that you had to design a 100 Mbps CSMA/CD protocol in which the maximum one-way propagation delay between any two hosts is 100×10^{-6} s. What will you use as the minimum size of a transmitted frame if you wish the transmitting node to detect a collision before completing the transmission of the frame (in Kilobits)

Your Answer: | Correct Answer: 20 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another is 40μ s. The minimum size of a frame in the network is ____ bytes.

Your Answer: | Correct Answer: 200 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame, collide, and A wins the first backoff race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second backoff race is:

A. 0.3125
B. 0.625
C. 0.75
D. 1.0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Suppose W is an even number. If packet $W/2$ is lost, what packet(s) will be retransmitted in GBN and what packet(s) will be retransmitted in SR?

A. GBN: $W/2, W/2+1, \ldots, W$ SR: $W/2+1$
B. GBN: $W/2+1, \ldots, W$ SR: $W/2+1$
C. GBN: $W/2, W/2+1, \ldots, W$ SR: $W/2$
D. GBN: $W/2+1, \ldots, W+1$ SR: $W/2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

In a sliding window ARQ scheme, the transmitter's window size is N and the receiver's window size is M. The minimum number of distinct sequence numbers required to ensure correct operation of the ARQ scheme is

A. $\min(M, N)$
B. $\max(M, N)$
C. MN
D. $M+N$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that only one station transmits in a given time slot?

A. $np(1-p)^{n-1}$
B. $(1-p)^{n-1}$
C. $p(1-p)^{n-1}$
D. $1-(1-p)^{n-1}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Identify and mark the correct statements below:

A. With the SR protocol, it is possible for the sender to receive an ACK for a packet that falls outside of its current window.
B. With GBN, it is possible for the sender to receive an ACK for a packet that falls outside of its current window.
C. The alternating-bit protocol is the same as the SR protocol with a sender and receiver window size of 1 .
D. The alternating-bit protocol is the same as the GBN protocol with a sender and receiver window size of 1 .

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

The maximum window size for data transmission using the selective reject protocol with n-bit frame sequence numbers is:

A. 2^n
B. 2^{n-1}
C. $2^n - 1$
D. 2^{n-2}

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25 ms . Frames are to be transmitted into this link to maximally pack them in transit (within the link). What is the minimum number of bits (I) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Suppose that it takes 1 unit of time to transmit a packet (of fixed size) on a communication link. The link layer uses a window flow control protocol with a window size of N packets. Each packet causes an ack or a nak to be generated by the receiver, and ack/nak transmission times are negligible. Further, the round trip time on the link is equal to N units. Consider time $i > N$. If only acks have been received till time i (no naks), then the goodput evaluated at the transmitter at time i (in packets per unit time) is

A. $1 - \frac{N}{i}$
B. $\frac{i}{(N+i)}$
C. 1
D. $1 - e^{\left(\frac{i}{N}\right)}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 05sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Consider slotted ALOHA. We can say a slot is "not wasted" whenever -

A. No station transmits.
B. Only one station transmits.
C. Multiple stations transmit.
D. None of these.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

When routers generate ICMP messages, to where do they send them?

A. The messages are sent to the source address specified in the IP header.
B. The messages are sent to the destination address specified in the IP header.
C. The messages are sent to next hop.
D. The messages are sent to previous hop.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Consider a network using the pure ALOHA medium access control protocol, where each frame is of length $1,000$ bits and the channel transmission rate is 1 Mbps . The aggregate number of transmissions across all nodes (including new and retransmitted frames) is modeled as a Poisson process with rate $1,000$ frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer in terms of frames per second) ____.

Your Answer: | Correct Answer: 136 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Let k be minimum number of message bits, in order to correct all possible t bits error then which of the following condition must satisfy?(n denotes the length of the codeword in bits)

A. $2^{n-k} \geq \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{t}$
B. $2^{n} \geq 1 + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{t}$
C. $2^{n-k} \geq 1 + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{t}$
D. $2^{k} \geq 1 + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{t}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Two CSMA/CD stations are each trying to transmit long (multiframe) files. After each frame is sent, they contend for the channel using the binary exponential backoff algorithm.
$i = 1$ is their first attempt, and results in the first collision. On attempt $i = 2$ each randomly skips either 0 or 1 slot. On the ith attempt each skips $0, 1, \ldots, 2^{i-1}$ slots randomly.

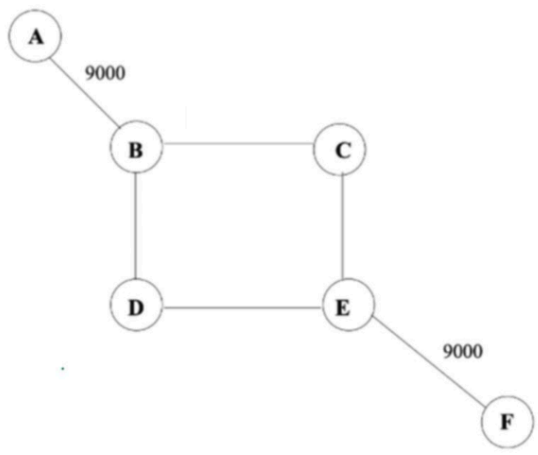
What is the probability that the contention ends on round $i = k$?

A. $\left(\prod_{i=1}^{k-1} \frac{1}{2^{i-1}}\right)\left(1 - \frac{1}{2^{k-1}}\right)$
B. $\left(\prod_{i=1}^{k-1} \frac{1}{2^{i-1}}\right)\left(1 - \frac{1}{2^{k}}\right)$
C. $\left(\prod_{i=1}^{k} \frac{1}{2^{i-1}}\right)\left(1 - \frac{1}{2^{k-1}}\right)$
D. $\left(\prod_{i=1}^{k-1} \frac{1}{2^{i}}\right)\left(1 - \frac{1}{2^{k-1}}\right)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

The figure shows a network consisting of six nodes. The links between the nodes have different MTUs. The MTU for the links $\mathrm{A} - \mathrm{B}$ and E-F is 9000 bytes. Node A sends an IP packet of size 2604 bytes to F. The packet undergoes fragmentation in the network, and F receives three fragments of size 1500 bytes, 572 bytes, and 572 bytes (not necessarily in that order). The sizes include the sizes of both the IP header and the payload.



A. B--C: 1500B
C--E: 1500B
B--D: 576B
D--E: 576B

B. B--C: 1500B
C--E: 1500B
B--D: 1500B
D--E: 576B

C. B--C: 1500B
B--D: 1500B
C--E: 576B
D--E: 576B

D. B--C: 576B
C--E: 576B
B--D: 576B
D--E: 576B

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

The distance between two stations M and N is L kilometers. All frames are K bits long. The propagation delay per kilometer is t seconds. Let R bits/ second be the channel capacity. Assuming that the processing delay is negligible, the minimum number of bits for the sequence number field in a frame for maximum utilization, when the sliding window protocol is used, is:

A. $\left\lceil \log_2 \frac{2LtR+2K}{K} \right\rceil$
B. $\left\lceil \log_2 \frac{2LtR}{K} \right\rceil$
C. $\left\lceil \log_2 \frac{2LtR+K}{K} \right\rceil$
D. $\left\lceil \log_2 \frac{2LtR+2K}{2K} \right\rceil$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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